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# How Debts Fall: Five Episodes of Sovereign Debt Reduction and the Strategies That Drove Them

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**Abstract.** We study five episodes in which governments carried very high public debt and brought it down substantially: Britain 1816 to 1914, the United States 1946 to 1974, Sweden 1991 to 1998, Canada 1993 to 2006, and Belgium 1993 to 2007. The five are chosen for maximum contrast in strategy, era, and institutional setting. Declines range from 35 to 213 percentage points of GDP. Three conditions appear in all five: a binding constraint on the soft options (inflation or delay), primary surpluses sustained over many years or a financial system the state could repress, and a forcing crisis that created political permission. External anchors, gold, Bretton Woods, euro qualification, or legislated fiscal rules, appear in four of five. The cleanest measured decomposition, for the United States, attributes 17 points of the 83-point decline to primary surpluses, 28 to interest-rate distortions, and 6 to their interaction (Reconstructed, A). No case reduced debt through growth alone, and the red-flag cases, Japan and Italy, show what happens when the checklist conditions fail.

**Keywords:** sovereign debt, fiscal consolidation, debt dynamics, financial repression, fiscal rules

**JEL Classification:** E62, E63, H63, N10, N40

## Non-technical summary

Governments across the advanced economies owe more today than they have in decades, and the question of how that debt comes down without breaking the economy is no longer academic. This paper studies five times it actually happened: Britain after the Napoleonic Wars, the United States after the Second World War, Sweden and Canada in the 1990s, and Belgium in the run-up to the euro. The methods were completely different. Britain ran budget surpluses for nearly a century. The United States kept interest rates artificially low and let inflation quietly shrink what it owed. Sweden, Canada, and Belgium cut spending, raised some taxes, and rewrote their budget rules.

Three conditions appear in all five cases. First, the easy ways out were blocked: no printing money, no postponing the reckoning. Second, the government either ran sustained surpluses or controlled interest rates directly; the arithmetic requires one of the two. Third, a crisis gave politicians the cover to act, and the rules they built kept their successors from reversing course. No case in this paper reduced debt through economic growth alone. The two failures we study show what happens when the conditions are missing: Japan never generated the surpluses, so its debt kept climbing past 230 percent of GDP, and Italy stopped consolidating once the pressure of euro qualification faded, so its gains reversed.

The practical read is narrow. Watch the gap between interest rates and growth, not just the deficit. Pick the method that fits the institutions you have. And expect the discipline to decay the moment the crisis passes.

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# 1 Introduction

Sovereign debt ratios are high across the advanced economies, and the question of how they come down is no longer academic. This paper studies five historical episodes in which governments carried very high public debt and brought it down substantially, chosen for maximum contrast in strategy, era, and institutional setting: Britain after the Napoleonic Wars, the United States after the Second World War, Sweden in the 1990s, Canada in the 1990s, and Belgium in the run-up to and early years of the euro.

The episodes span two centuries and five genuinely different playbooks. Britain ran primary surpluses for the better part of a century under the gold standard and reduced debt from about 240 percent of GDP to about 27 percent (Observed, B). The United States used financial repression, a pegged yield curve, surprise inflation, and retained wartime taxes to take debt held by the public from 106 percent to 23 percent in 28 years (Observed, A). Sweden rebuilt its fiscal institutions after a triple crisis, moving from an 11 percent deficit to surplus in five years (Observed, B). Canada cut program spending about 10 percent in nominal terms in two years under the 1995 Program Review and halved its federal debt ratio in a decade (Observed, B). Belgium ran sustained primary surpluses under the external discipline of euro qualification, taking gross debt from about 134 percent to about 84 percent (Observed, B).

Three findings cut across the cases. First, every reduction combined a binding constraint on the soft options with either sustained primary surpluses or a financial system the state could repress; no case succeeded on growth alone. Second, the sign of the interest-growth differential determined the strategy: Britain consolidated against adverse arithmetic, the United States and Belgium engineered favorable arithmetic, and Belgium's decade of surpluses before 1993 achieved nothing until euro convergence flipped the sign. Third, the output cost of consolidation depended less on speed than on the starting cyclical position: it was lowest where the recession preceded the pivot.

The rest of the paper proceeds as follows. Section 2 sets out the debt-dynamics identity that organizes the analysis. Section 3 reviews the related literature. Section 4 describes the data and Section 5 the methodology. Section 6 presents the five cases in condensed form and then the comparative anatomy: external anchors, repression versus surpluses, the democracy question, the speed-versus-cost tradeoff, a tiered checklist of observable conditions, and the red-flag cases. Section 7 states the limitations and what would falsify the conclusions. Section 8 concludes. A data appendix follows the references.

# 2 Institutional background: the debt-dynamics identity

The change in the debt-to-GDP ratio follows an accounting identity. Let $d_t$ be the debt ratio, $p_t$ the primary deficit as a share of GDP, $r_t$ the real interest rate on the debt, $g_t$ the real growth rate, and $sfa_t$ the stock-flow adjustment (privatization receipts, valuation changes, and other below-the-line items). Then, to a close approximation,

$$d_t - d_{t-1} = p_t + (r_t - g_t)\, d_{t-1} + sfa_t. \tag{1}$$

Three terms move the ratio: the primary balance, the interest-growth differential applied to the inherited stock, and one-off stock adjustments. Everything a government does to reduce debt operates through one of these channels. Running a primary surplus is $p_t < 0$. Growing faster than

the interest rate is $r_t - g_t < 0$. Selling assets or inflating away nominal liabilities shows up in $sfa_t$ or in a depressed $r_t$.

The identity disciplines the case studies. A government that runs primary surpluses while $r > g$ is pushing against adverse arithmetic, which is what Britain did for decades. A government that holds $r$ below $g$ through rate pegs, captive domestic savers, or surprise inflation is practicing financial repression, which is what the United States did after 1945 (Reinhart and Sbrancia, 2015). A government that sells gold reserves or telecoms is using $sfa_t$, which is what Belgium did at the margin. No strategy escapes the identity; the question in each case is which terms did the work.

## 3 Related literature

This paper sits at the intersection of three literatures. The first is the historical literature on debt reduction episodes. Reinhart and Rogoff (2011) document the long arc from financial crash to debt crisis across eight centuries and establish that high debt ratios have repeatedly been followed by long consolidations. Acalin and Ball (2024) provide the cleanest quantitative decomposition of the postwar US episode, constructing counterfactual undistorted real rates and attributing the 1946 to 1974 decline across primary surpluses, interest-rate distortions, and their interaction; their numbers anchor Section 6.2 below.

The second is the literature on financial repression. Reinhart and Sbrancia (2015) define the repression toolkit, negative real rates, captive domestic audiences, directed credit, and capital controls, and argue it was the dominant channel of postwar debt liquidation across the advanced economies. Sbrancia (2011) quantifies the cumulative liquidation effect of inflation for a panel of countries from 1945 to 1980, finding, for example, a 40 percentage point reduction for the United States relative to a constant 2 percent inflation counterfactual.

The third is the literature on fiscal adjustments. Alesina and Ardagna (2010) distinguish tax-based from spending-based consolidations and find the latter less costly in output terms, a distinction visible in the Canada and Sweden cases below. Blanchard (2019) and Furman and Summers (2020) reopened the question of whether $r < g$ makes debt self-stabilizing; Elmendorf and Mankiw (1999) summarize the older consensus that debt imposes burdens through crowding out. Acalin and Ball (2024) speak directly to the Blanchard debate: strip out the postwar distortions, and the US debt ratio falls only to 74 percent by 1974, not 23 percent, which implies the benign postwar arithmetic was substantially engineered rather than natural.

## 4 Data

The paper draws on official and near-official sources for each case: the Congressional Research Service and OMB Historical Tables for US debt ratios; IMF Article IV consultations for Sweden (1999) and Belgium (1996); the European Commission's 2011 peer review and Belgium's stability programmes; the Riksbank for Swedish monetary history; the Bank of England's historical macroeconomic dataset as compiled by ukpublicspending.co.uk for Britain; and Department of Finance Canada Public Accounts as compiled by the Fraser Institute and the Mercatus Center for Canada. Secondary compilations fill gaps where primary series do not reach: Martin Wolf in the Financial Times for Britain's 1816 peak, Acalin and Ball's IMF working paper for the US decomposition, and Bisciari et

al. (2015) via MDPI for Belgium's. Italy's counter-case draws on the OFCE Sciences Po policy brief 56/2019.

Table 1 summarizes the five episodes. Definitions differ across two centuries: Britain's figure is net public debt, the US figure is debt held by the public, and the modern European cases are general government gross debt. The direction and rough magnitude of each decline do not hinge on these definitional differences; exact comparability is not claimed.

**Table 1.** The five episodes: peak, trough, and decline

| Case | Peak year | Peak debt/GDP | Trough year | Trough debt/GDP | Decline (pp) | Years |
|---|---|---|---|---|---|---|
| Britain | 1816 | 240 | 1914 | 27 | 213 | 98 |
| United States | 1946 | 106 | 1974 | 23 | 83 | 28 |
| Sweden | 1995 | 76 | 2015 | 29 | 47 | 20 |
| Canada | 1993 | 67 | 2006 | 32 | 35 | 13 |
| Belgium | 1993 | 134 | 2007 | 84 | 50 | 14 |

Notes: Debt definitions differ by case (see Section 4): Britain is net public debt, the US is debt held by the public, the rest are general government gross debt (Canada is the federal series). Peak and trough years are approximate; decline is in percentage points of GDP.
Source: Memo compilation from CRS/OMB, IMF Article IVs, Riksbank, EC peer review, Fraser Institute, ukpublicspending.co.uk.

# 5 Methodology

The analysis is comparative case study, organized by the debt-dynamics identity of Section 2. For each case we report the peak and trough ratios with evidence grades, describe the strategy through the three channels of the identity, attribute the decline across channels where published decompositions exist, and document the political economy, the costs, and the near-failures. Where the memo attributes points of decline to components, the figures come from the cited authors' own decompositions, not from recomputation here.

Three methodological choices deserve stating. First, we report ratios, not nominal stocks, because the ratio is the burden measure; but we note where the nominal debt actually fell (Canada, Belgium's near-flat numerator) versus where only the denominator moved (Britain's nominal debt was barely reduced for decades). Second, we grade every key number with the evidence vocabulary in the paper standards: Observed, Reconstructed, Modelled, or Assumed, with source-quality grades A through C. Third, this is a retrospective analysis; there was no pre-registered plan, and the five cases were selected for strategy contrast, which means the sample is chosen on the outcome. The checklist in Section 6.6 is therefore a description of what the successes shared, not a tested forecasting model.

**Production note.** This paper was produced by Kilambi Research in October 2026. The five case studies were researched from primary sources (IMF Article IV documents, central bank histories, national fiscal accounts, and official stability programmes) and compiled in a research memo; figures were rendered with Kilambi Research's institutional chart style. Three load-bearing figures were independently verified before publication: the Acalin-Ball US decomposition, Sweden's 500 percent rate, and Britain's 1816 debt peak (see Section 7). Data were retrieved in October 2026 and are

subject to revision by the underlying sources. All analysis was performed by Kilambi Research's quantitative research pipeline with AI assistance and verified against source data.

# 6 Results

## 6.1 Britain, 1816 to 1914: the century-long grind

British net public debt reached about 240 percent of GDP in 1816, the fiscal legacy of 125 years of war against France (Observed, B); compilations range from 237 percent to over 250 percent depending on definition and year, and some sources simply say more than 200 percent, so 240 percent is the standard figure with the range disclosed. By 1914 the ratio had fallen to about 25 to 30 percent (Observed, B). The consolidation took roughly a century.

The strategy was the simplest in this paper and the hardest to copy: run primary surpluses, almost every year, for the better part of a century, under a hard-money regime. The government demobilized, withdrew occupation forces from France after 1818, and brought the budget into small surplus. Peel reintroduced the income tax permanently in 1842 and used it to cut customs and excise while keeping revenue up. Gladstone made balanced budgets a moral doctrine: surpluses existed only to reduce the debt. The gold standard, formally resumed in 1821, removed the inflation escape route entirely. The Treasury also actively cut the interest bill, converting 135 million pounds of 5 percent bonds into 4 percents in 1823 and 80 million of 4 percents into 3.5 percents in 1824 (Observed, B).

The striking fact is that the nominal debt was barely reduced at all for decades; the ratio collapsed because the denominator grew. But this was not painless growth. Consols yielded 3 to 4 percent while real growth ran about 2 percent in the first half of the century (Observed, B), so with $r > g$ the arithmetic required primary surpluses just to stand still, and the surpluses were delivered: debt interest absorbed close to half of all UK public spending between 1815 and 1855 (Observed, B). The Victorian growth spurt of the third quarter of the century then did the heavy lifting on the denominator.

The politics were pre-democratic for the decisive decades: the franchise was unreformed before 1832 and narrow until 1867 and 1884, and the spending constituencies that make modern consolidation agonizing, welfare, pensions, health, did not exist. The forcing crisis was the debt itself plus the postwar distress of the 1810s and 1820s. The 1825 panic, Britain's first modern financial crisis, followed the aggressive debt conversions and forced a restructuring of the financial system; the Chartist agitation of 1848 threatened the fiscal order politically. The famous punchline, from Wolf: with this crushing burden of debt, what economic disaster followed? The industrial revolution.

## 6.2 United States, 1946 to 1974: financial repression and the stop-digging rule

Gross federal debt stood at 118.9 percent of GDP in fiscal 1946; debt held by the public, the cleaner measure, was about 106 to 109 percent (Observed, A). By 1974 debt held by the public had fallen to about 23 percent (Observed, A). The ratio fell roughly 83 percentage points in 28 years.

The United States did not austerity its way out. It used two distortions deliberately and one tailwind gratefully. First, the Federal Reserve pegged Treasury yields from 1942 until the 1951 Treasury-Fed Accord: 0.375 percent on bills, 2.5 percent on the long bond (Observed, A). Second, after the

peg ended, Regulation Q capped deposit rates, capital controls bottled up domestic savings, and inflation repeatedly surprised to the upside; real interest rates were negative about half the time between 1945 and 1980 (Observed, B). Third, the fiscal stance followed the "stop digging" rule: the nominal debt was essentially not allowed to rise, war taxes were retained, military spending collapsed with demobilization, and 2.6 percent real growth with 3.3 percent inflation ground the ratio down to 46 percent by 1960 (Observed, B).

The decomposition is the best measured in this paper. Acalin and Ball (2024) construct counterfactual undistorted real rates and find that of the 83-point fall, 51 points are explained by the combination of primary surpluses and interest-rate distortions: 17 points from surpluses alone, 28 from distortions alone, and 6 from their interaction (Reconstructed, A). Strip out both, leaving only undistorted growth, and debt falls to just 74 percent by 1974, then drifts upward. An earlier draft of this paper's source memo reported a 30/40 split from a secondary summary; the figures here are corrected to the working paper's own text, verified before publication. Figure 2 shows the decomposition. The lesson is quantitative: repression did more work than fiscal virtue, 28 points to 17, with undistorted growth contributing 32. Table 2 reports the full decomposition.

**Table 2.** Decomposition of the US debt-ratio decline, 1946 to 1974

| Component | Points of GDP | Share of decline (%) |
|---|---|---|
| Primary surpluses alone | 17 | 20 |
| Interest-rate distortions alone | 28 | 34 |
| Interaction of surpluses and distortions | 6 | 7 |
| Undistorted growth | 32 | 39 |
| Total decline | 83 | 100 |

Notes: Components sum to the 83-point fall in debt held by the public, from about 106 percent of GDP in 1946 to about 23 percent in 1974. Figures are (Reconstructed, A), taken from the counterfactual decomposition in Acalin and Ball (2024), IMF Working Paper 24/5. Shares are rounded.

Source: Acalin and Ball (2024), verified against the working paper's text before publication.

The politics were unusually permissive: wartime tax rates were kept without any politician voting for an increase, demobilization cut spending automatically, and Bretton Woods created captive global demand for Treasuries. The costs fell on savers and bondholders through negative real returns, not on workers: unemployment stayed low and the period covers the golden age of American productivity growth. The peg nearly broke in 1950 to 1951, resolved by the Treasury-Fed Accord; by the late 1960s, guns-and-butter spending ended the surplus era, and the 1970s delivered the uncontrolled version of what the 1940s and 1950s had managed in controlled form.

### 6.3 Sweden, 1991 to 1998: the crisis that rebuilt the fiscal state

Sweden's general government gross debt peaked at about 73 to 78 percent of GDP in the mid-1990s (Observed, B); the fiscal deficit hit 11.3 to 12 percent of GDP in 1993 (Observed, B). By 1998 the budget was in surplus at 2.2 percent of GDP (Observed, B), and gross debt fell to about 45 percent by 2007 and below 30 percent by 2015 (Observed, B). The deficit-to-surplus turn took roughly five years.

The strategy was institutional reconstruction. The Bildt center-right government began with the

banking rescue: a blanket guarantee in September 1992, nationalization of insolvent banks, and bad banks at a fiscal cost of 2.3 percent of GDP in 1993 and 1.5 percent in 1994 (Observed, B). The Persson Social Democratic government then delivered a consolidation worth 7.5 percent of GDP over 1994 to 1998, roughly 80 percent spending cuts and 20 percent tax increases in the early years (Observed, B). The permanent legacy was the framework: rolling three-year expenditure ceilings from 1996, a 2 percent surplus target over the cycle from 1997, an independent Riksbank with a 2 percent inflation target from 1993, and a notional-defined-contribution pension reform in 1998. Monetary policy helped from the other side: after the krona was floated on November 19, 1992, following a failed defense in which the Riksbank raised its marginal rate to 500 percent on September 16, 1992 (Observed, A), the 20 percent depreciation powered an export-led recovery with GDP growing 4.1 percent in 1994.

The IMF's 1999 Article IV called the structural improvement "10 percent of GDP," from spending cuts and tax increases augmented by reduced interest costs (Observed, B). Public expenditure fell from 67 percent of GDP in 1993 to about 52 percent by 2015 (Observed, B). Unlike the US case there was no financial repression and no inflation surprise: inflation stayed near or below target throughout. This was the cleanest primary-surplus-driven consolidation in the paper.

The remarkable feature is cross-party ownership: the party that built the welfare state deepened the cuts. The forcing crisis was total: banking, currency, and fiscal crises simultaneously, GDP down 5 percent cumulatively, unemployment jumping from 1.5 to 8 percent in two years. The costs were front-loaded and severe, but growth recovered strongly from 1994, and the fiscal space created has since funded defense and energy commitments without borrowing stress. Two moments nearly broke it: the currency defense itself, which deepened the banking crisis it meant to contain, and the 1994 election, which could have reversed the consolidation as ideological betrayal. Persson doubled down instead.

### 6.4 Canada, 1993 to 2006: Program Review

Federal government debt peaked at about 67 to 68 percent of GDP in 1993 to 1996 (Observed, B); general government net debt including provinces peaked near 100 percent around 1996 (Observed, B). By 2006 to 2008 federal debt was about 32 percent of GDP, and the government had paid down 100 billion Canadian dollars, 15.2 percent, of the nominal national debt between 1996-97 and 2007-08 (Observed, B). The deficit-to-surplus turn took three years; the debt-ratio halving took about a decade.

The strategy was expenditure-led consolidation at unusual speed and depth. The 1994 Program Review, a zero-based review of every federal program, laid the groundwork for Finance Minister Paul Martin's landmark 1995 budget under Prime Minister Chretien. Program spending was cut about 10 percent in nominal terms over two years and about 14 percent in real terms (Observed, B), deliberately spending-heavy at six to seven dollars of cuts per dollar of tax increases (Observed, B). Federal program spending fell from 17.5 percent of GDP in 1992-93 to 11.3 percent in 2000-01, the lowest in more than half a century (Observed, B). The deficit went from 36.6 billion dollars (4.6 percent of GDP) in 1994-95 to a 3 billion surplus in 1997-98, the first since 1969-70 (Observed, B).

This was overwhelmingly a primary-surplus story. Debt-service costs fell from 5.6 to 4.3 percent of GDP as credibility returned (Observed, B); growth was solid, aided by US demand and a competitive currency; there was no inflation surprise and no financial repression. Nominal debt actually fell, which is rarer than it sounds.

The politics are the most interesting after Britain's: the consolidation was executed by Liberals, the center-left party, which inoculated it against partisan attack. The forcing crisis was the 1994 Mexican peso crisis and the bond market's verdict that Canada had become "an honorary member of the Third World" (Observed, B); by 1995 the federal government was paying more than one of every three revenue dollars in interest alone (Observed, B). The October 1995 Quebec sovereignty referendum, decided by about one percentage point, could have made the fiscal program irrelevant beside national breakup. The Chretien government then ran surpluses for a decade. Costs were concentrated in transfers to provinces, downloaded to hospitals, universities, and welfare rolls, but there was no deep recession: growth continued through the cuts.

### 6.5 Belgium, 1993 to 2007: the euro anchor

General government gross debt peaked in 1993 at about 131 to 138 percent of GDP (Observed, B); by 2007 it had fallen to about 84 to 85 percent (Observed, B). Roughly 50 percentage points in under 15 years, the largest peacetime reduction in continental Europe in the period.

The strategy was externally anchored consolidation sustained over an unusually long period. The Maastricht convergence criteria gave every Belgian government from 1992 a non-negotiable external target: qualify for the euro. The Dehaene governments' 1993 Global Plan combined expenditure restraint (growth norms on social security, contained benefit indexation, restricted unemployment access, postponed retirement, higher healthcare co-payments) with heavy revenue measures: a 3 percent crisis surcharge on income taxes, de-indexed brackets, higher VAT and excises. Between 1992 and 1998 state revenues rose by 5 percentage points of GDP (Observed, B). Asset sales, gold reserves, public credit institutions, Belgacom, deleveraged the stock directly. Remarkably, Belgium had run a primary surplus since 1983 (Observed, B): the problem before 1993 was that interest rates exceeded growth by enough to overwhelm it. Euro-area integration then structurally reduced the interest burden from 1993 onward, flipping the $r - g$ arithmetic.

The decomposition, from Bisciari et al. (2015): after the 1993 peak, nominal debt grew by merely 18 percent while nominal GDP surged 81.2 percent (Reconstructed, B). Roughly speaking, the denominator did about four-fifths of the work while primary surpluses held the numerator nearly flat; Figure 2 shows this approximate split. Falling interest costs, as markets priced in euro membership, were the swing factor: they converted a primary surplus that had been insufficient since 1983 into one that cut the ratio by more than 3 points a year. No inflation surprise, no classic financial repression; the "repression" was the convergence trade itself compressing Belgian spreads.

Belgian politics is famously fragmented along linguistic lines; the Maastricht criteria provided an external, non-partisan target no coalition partner could openly oppose without opposing Europe itself. Budgetary targets were embedded in coalition agreements with strong monitoring by the Budget Minister and independent forecasts from the Federal Planning Bureau (Observed, B). Costs were fiscal rather than macroeconomic: the tax burden rose further, benefit indexation was contained, and a 1993 general strike protested the Global Plan, but growth continued and unemployment did not spike. The anchor decayed after qualification: during 2002 to 2004 the primary surplus slipped about 1 point of GDP as interest savings were spent (Observed, B), and after 2008 the debt rose again.

### 6.6 Comparative anatomy

Table 3 compares the five cases on the dimensions the analysis uses: the scale and speed of the decline, the arithmetic engine that did the work, the external anchor, the sign of the interest-growth differential, and the output cost.

**Table 3.** The five consolidations compared

| Case | Window | Decline | Primary engine | External anchor | $r - g$ |
|---|---|---|---|---|---|
| Britain | 1816–1914 | 213 pp / 98 yrs | Primary surpluses | Gold standard | Adverse |
| United States | 1946–1974 | 83 pp / 28 yrs | Financial repression | Bretton Woods | Favorable* |
| Sweden | 1991–1998 | 47 pp / 20 yrs | Primary surpluses | EU + fiscal framework | Neutral |
| Canada | 1993–2006 | 35 pp / 13 yrs | Primary surpluses | Market-imposed | Favorable |
| Belgium | 1993–2007 | 50 pp / 14 yrs | Primary surpluses | Maastricht/euro | Flipped† |

Notes: Decline in percentage points of GDP from peak to trough; window lengths differ by case (see Table 1). *Favorable by design: the United States engineered low real rates through the peg, Regulation Q, and surprise inflation. †Belgium ran primary surpluses from 1983 with no result until euro convergence flipped $r - g$ from adverse to favorable after 1993. Engine classifications summarize Sections 6.1–6.5.
Source: Author compilation from the case studies above.

**External anchors.** Four of the five cases used an external anchor to make the consolidation commitment credible, and the fifth invented a domestic equivalent. Britain had the gold standard: it removed the inflation option by law and convention. The United States had Bretton Woods and the dollar's reserve role: captive global demand for Treasuries made the peg and the repression sustainable. Belgium had the Maastricht criteria and euro qualification: a dated, binary, externally judged target. Sweden had EU accession plus, more importantly, the domestically legislated fiscal framework, which functioned as a self-imposed external anchor. Canada is the partial exception: no formal external anchor, but the peso crisis and the bond market's verdict acted as a market-imposed one. The mechanism matches the commitment-device finding of our companion disinflation paper: the anchor makes reversal costly and expectations stable.

**Repression versus surpluses.** The five cases span the full spectrum. The United States is the repression-dominant case: 28 of the 83 points of decline came from interest-rate distortions against 17 from primary surpluses. Britain is the surplus-dominant case under adverse arithmetic: with $r > g$, surpluses were mathematically necessary and were delivered for decades. Sweden, Canada, and Belgium are the modern institutional cases: primary surpluses did the work, with growth and falling interest rates as accomplices, and no inflation surprise. The summary rule: repression works when the state controls the financial system and the currency is dominant; surpluses work when institutions bind governments across elections. No case in this paper reduced debt through growth alone. Figure 4 shows the approximate $r - g$ sign for each consolidation window: adverse only in Britain, favorable or neutral elsewhere.

**Did democracy matter.** All five consolidations happened under democratic government, which is worth stating against the claim that only autocracies can impose fiscal pain. But the selection carries a large caveat: Britain did most of its consolidating before mass democracy, and the modern cases all used the same political technology, a crisis that created permission plus institutional rules that bound successors. Canada and Sweden add the left-government paradox: center-left parties

executing austerity inoculate against partisan reversal. None of the five is an autocracy, and none needed to be.

**Speed versus output cost.** The tradeoff is visible but not linear, as Figure 3 shows. Britain: a century, minimal annual pain, maximal cumulative interest burden. The United States: 28 years, minimal pain, costs shifted to bondholders. Belgium: 14 years, modest pain, costs shifted to taxpayers. Canada: a decade to halve the federal ratio, modest pain concentrated in transfers. Sweden: five years to turn the deficit around, but the consolidation followed rather than caused a deep recession, so the marginal pain of the tightening itself was arguably the smallest. The generalization: the output cost of consolidation is lowest when the recession precedes the pivot and highest when consolidation must create the surplus from a standing start against adverse $r - g$. Speed itself did not determine cost; the starting cyclical position did.

**Checklist: observable conditions for successful large-scale debt reduction.** Tier 1, present in all five: (1) a binding constraint on the soft options, gold, reserve-currency regime, euro-qualification criteria, legislated fiscal rules, or market-imposed funding pressure; (2) primary surpluses sustained over many years, or a financial system the state can repress, one of the two being arithmetically necessary; (3) a forcing crisis that discredits the old fiscal regime and creates political permission. Tier 2, present in four of five: (4) an external anchor with a date or a judge; (5) cross-party or cross-coalition ownership so elections do not reverse the consolidation; (6) a growth tailwind during the consolidation, from technological revolution, postwar catch-up, currency depreciation, external demand, or convergence. Tier 3, judgment: (7) watch $r - g$, not just the deficit, Belgium consolidated for a decade before 1993 with primary surpluses and got nowhere until convergence flipped the sign; (8) nominal debt stabilization as an explicit target, since ratio-only targets invite denominator gambling.

**Red flags: where high debt did not fall.** Japan is the clean negative case. Public debt crossed 100 percent of GDP in 1997 and kept rising past 230 percent (Observed, B). Every Tier 1 condition failed in the opposite direction: persistent primary deficits, deflation instead of inflation (which raises the real debt burden), no forcing crisis sufficient to create permission (the 1990s banking crisis was managed with forbearance, not resolution, unlike Sweden's), and no binding constraint (Bank of Japan bond buying removed market pressure rather than imposing it). Japan shows that low interest rates alone do not reduce debt: without primary surpluses, the ratio rises no matter how low $r$ goes if $g$ is lower still and the deficit persists.

Italy is the subtler negative case, and the more instructive one, because Italy initially succeeded. The 1992 Amato budget began a consolidation that lasted over a decade: the debt ratio fell from a 117 percent peak in 1997 to below 100 percent in 2007 (Observed, B). Then it reversed. The euro anchor decayed after qualification, growth never arrived, and after 2008 the ratio climbed back above 130 percent (Observed, B). Italy's lesson for the checklist: Tier 2 condition 6, the growth tailwind, is load-bearing. Consolidation without subsequent growth is a holding action, not a resolution. Belgium kept consolidating after 1999 and held its gains until 2008; Italy stopped, and lost them.

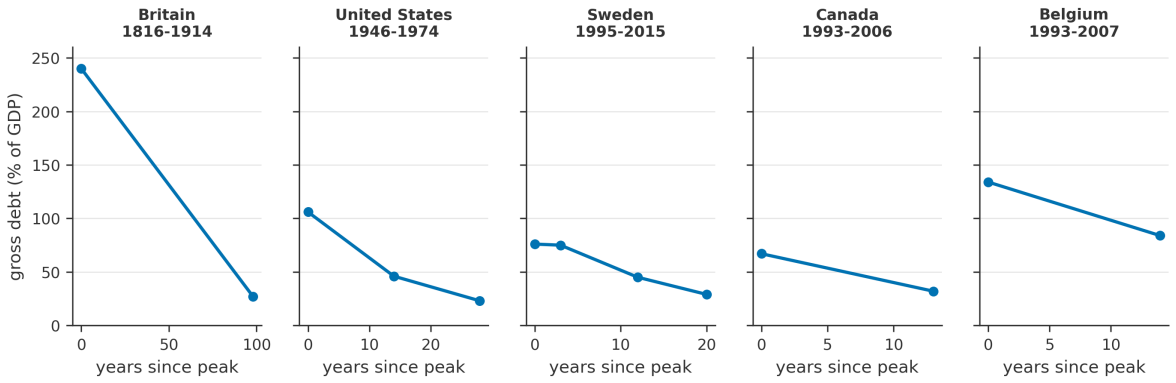



**Figure 1.** Debt-to-GDP paths for the five cases, indexed to years since the debt peak. Shared y-axis; x-ranges differ because Britain's consolidation spans a century. Only observed points are plotted; lines join adjacent observations.

# 7 Limitations and robustness

This is a retrospective, outcome-selected comparative study, and the limits are material. First, the five cases were chosen for strategy contrast among successes, so the checklist describes what successes shared; it is not a tested forecasting model, and the sample cannot establish which conditions are necessary. Second, definitions differ across cases and centuries: net versus gross, federal versus general government, and the 19th-century British figures rest on historical reconstructions, grade B throughout. Third, decompositions exist in published form only for the United States and approximately for Belgium; the attribution language for Britain, Sweden, and Canada is qualitative and should be read that way. Fourth, the $r - g$ bars in Figure 4 are desk estimates, grade C, included for the sign pattern. Fifth, the political-economy claims, cross-party ownership, crisis-created permission, rest on narrative evidence and cannot be falsified by the numbers alone.

Three load-bearing figures were independently verified before publication. The Acalin-Ball decomposition is 17 points primary surpluses, 28 points interest-rate distortions, and 6 points interaction, of the 83-point 1946 to 1974 fall; the figure is taken from the text of IMF Working Paper 24/5, confirmed in two independent mirrors of the paper. An earlier draft of this paper's source material reported a 30/40 split from a secondary summary; that split is corrected here. Sweden's 500 percent marginal rate is confirmed by the Riksbank's own history: September 16, 1992. Britain's 1816 peak of 240 percent of GDP is the standard figure from Martin Wolf in the Financial Times; other compilations range from 200 to 275 percent depending on definition and year, and the range is disclosed in the text.

**Falsifiability.** The checklist's Tier 1 claims would be weakened by a large, sustained, peacetime debt reduction achieved with no binding constraint on the soft options, no primary surpluses, and no repressible financial system, that is, by growth alone against neutral arithmetic. The speed-versus-cost claim would be falsified by a consolidation that created a deep recession from a standing start at full employment and still completed quickly. The repression-versus-surplus spectrum would be falsified by a clean published decomposition showing a major reduction driven primarily by undistorted $r < g$ with neither surpluses nor distortions. On current evidence, no such case is in the sample.

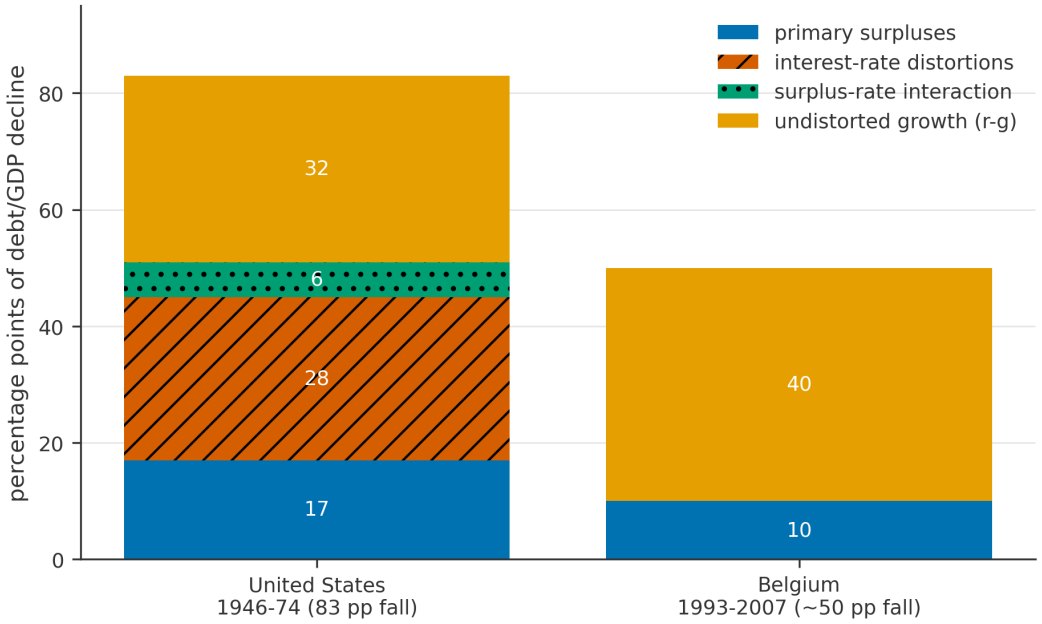



**Figure 2.** Decomposition of the debt-ratio decline. The US split is from Acalin and Ball (2024): 17 points primary surpluses, 28 points interest-rate distortions, 6 points interaction, 32 points undistorted growth, of the 83-point fall. The Belgium split is approximate, from nominal debt (+18%) versus nominal GDP (+81.2%) figures in Bisciari et al. (2015): the denominator did about four-fifths of the work. No published decomposition exists for Britain, Sweden, or Canada.

## 8 Conclusion

Five governments, two centuries, five different ways out from under. Britain ground its debt down over a century of surpluses under gold. The United States repressed its way out in 28 years with pegged rates and surprise inflation. Sweden rebuilt its fiscal institutions in five years after a triple crisis. Canada cut spending 10 percent in two years and halved its federal debt ratio in a decade. Belgium anchored 14 years of surpluses to euro qualification and took 50 points off the ratio. What they shared was not a tactic but a structure: a binding constraint, an arithmetic engine (surpluses or repression), and a crisis that made the politics possible. What the failures shared was the absence of that structure: Japan never generated the surpluses, and Italy stopped consolidating once the anchor decayed and growth never came. The practical read for today's high-debt sovereigns is correspondingly narrow: watch $r-g$, not just the deficit; pick the arithmetic that fits the institutions; and expect the anchor to decay the moment the prize is won.

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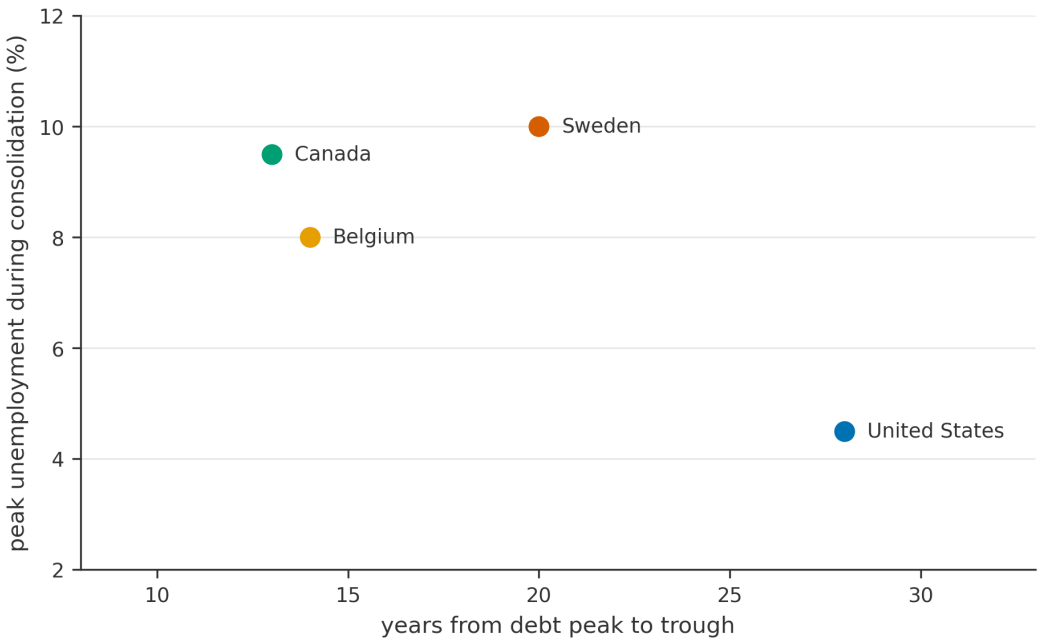



**Figure 3.** Speed versus output cost: years from debt peak to trough against peak unemployment during the consolidation. Britain is excluded; no unemployment series exists for the 19th century. The pattern is not a clean tradeoff: the US took 28 years with minimal unemployment, while Sweden's five-year turnaround followed a recession that had already done the damage.

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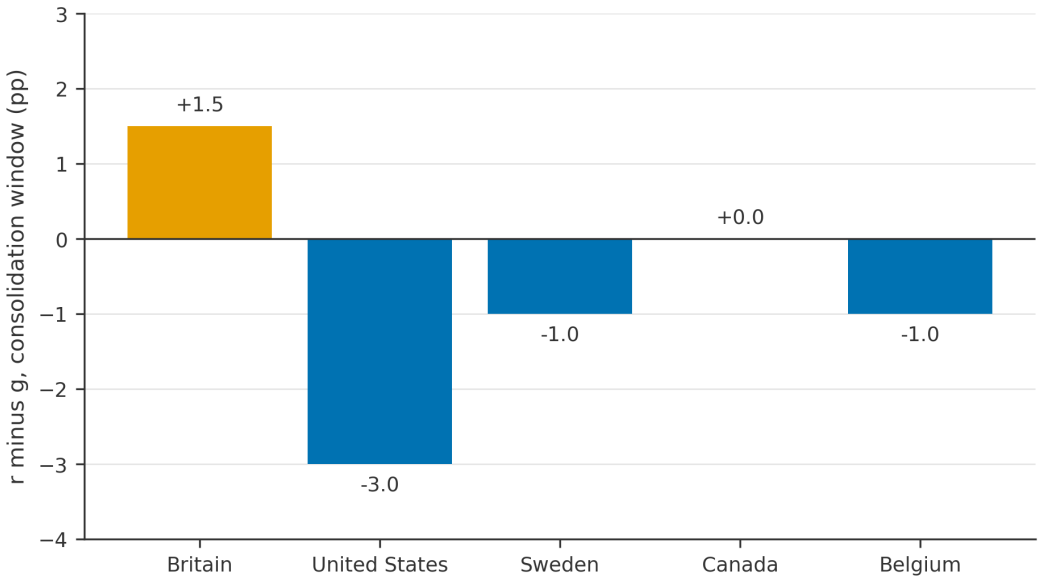



**Figure 4.** Approximate sign of $r - g$ during each consolidation window. Only Britain consolidated against adverse arithmetic ($r > g$), which is why its surpluses had to be sustained for a century. Estimates are (MODELLED, C); the sign, not the precision, is the point.

# A Data appendix

| Series | Definition | Source |
|---|---|---|
| Britain debt/GDP | Net public debt / GDP, 1816–1914 | ukpublicspending.co.uk, drawing on Bank of England historical data |
| US debt/GDP | Debt held by the public / GDP, FY1946–1974 | CRS R44383, OMB Historical Tables |
| US decomposition | Counterfactual attribution, 1946–1974 | Acalin and Ball (2024), IMF WP/24/5 |
| Sweden debt/GDP | General government gross debt / GDP | IMF 1999 Article IV; case digest; Reuters |
| Sweden deficit | General government net lending / GDP, 1993 | IMF 1999; Riksbank Governor Backstrom (2000) |
| Sweden 500% rate | Riksbank marginal rate, Sep 16 1992 | Sveriges Riksbank history |
| Canada debt/GDP | Federal debt / GDP, 1993–2006 | Fraser Institute; Mercatus Center, via Dept. of Finance Public Accounts |
| Belgium debt/GDP | General government gross debt / GDP | IMF 1996 Article IV; EC 2011 peer review; Belgium stability programme |
| Belgium decomposition | Nominal debt +18%, nominal GDP +81.2% | Bisciari et al. (2015), via MDPI |
| Italy debt/GDP | General government gross debt / GDP | OFCE Sciences Po policy brief 56/2019 |
| Japan debt/GDP | General government gross debt / GDP | Widely reported; IMF historical series |

Notes: All series retrieved October 2026. Historical reconstructions (Britain) and secondary compilations (Fraser, Mercatus) are graded B. The desk performed no independent recomputation of cited authors' decompositions.